

Experiment No: 05

Date : _____

Roll No: _____

Aim : To Draw Different Types of scale

Theory :

Scale :

- A **scale** is defined as the ratio of the linear dimensions of the object as represented in a **drawing** to the actual dimensions of the same.
- Some Drawing required to be drawn proportionally **reduced** or **enlarged**.
- The proportion by which we are enlarging or reducing actual length of the object on drawing is called as **scale**.

Scale Size :

1. Full size :

- The scale in which objects are drawn with actual measurement is called as full size scale.
- It is written in ration 1:1

2. Reduced size:

- The scale in which objects are drawn with reduced proportion is called as reduced size scale.
- It is written in ration 1:2 , 1:5, 1:10, 1:20

3. Enlarged size:

- The scale in which objects are drawn with increased proportion is called as reduced size scale.
- It is written in ration 2:1 , 5:1, 10:1, 20:1

Type Of Scale :

1. Plain Scale.
2. Diagonal Scale.

1. Plain Scale:

- This scale is used when very small distances are to be accurately measured for example 0.1 mm
- When measurements are required up to one decimal place (i.e. 2 unit)
- Example : 6.7 , 5.3

(Note : draw diagram on blank page)

Problem 1 :

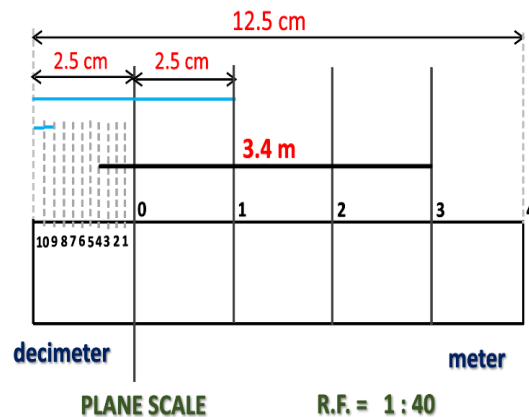
Construct a plain scale with R.F. 1:40 to read up to 5 meters in meters and decimeters. Mark 3.4 m on scale

Given data : R.F. = 1/40 , maximum length = 5 m

1. Calculate the length of scale.

$$\begin{aligned}\text{length of scale} &= \text{R.F.} \times \text{length to measured} \\ &= 1/40 \times 5 \text{ m} \\ &= 0.125 \text{ m} \\ &= 0.125 \times 100 = 12.5 \text{ cm}\end{aligned}$$

2. Draw two horizontal lines of 12.5 cm parallel to each other and 2 cm distance apart. Join the ends by vertical lines.



3. Divide horizontal lines into **required** number of **equal parts** (depend upon length to measured or read up to) according to given data for above example maximum length **or read up to is 5 m**, hence divide parallel lines into **5 equal parts**.
Size of individual part = $12.5/5 = 2.5 \text{ cm}$
4. Mark 'zero' at the end of first division.
5. Sub-divide only first main part to the left of zero into the **required** number of sub-division according to unit. In above problem given units are **meter** and **decimeter** and **1 meter = 10 decimeter** hence we have to divide first main part into **10** number of **equal parts**.
Size of individual sub part = $2.5/10 = 0.25 \text{ cm}$
6. Sub-divisions are numbered to the right, starting from zero.
7. Assign units for parts.
8. Mark given distance on scale. For above problem distance = **3.4 m**
9. Name of the scale and R.F. should be mentioned below scale

2. Diagonal Scale.

- Diagonal scale is used in engineering to read lengths with higher accuracy as it represents into three different unit for example meters, centimeters and millimeters.
- When measurements are required up to two decimal place (i.e. 3 unit)
- Example : 6.71 , 7.34

Problem :

Construct a diagonal scale of **1:50** to read **meters, decimeters and centimeters** to measure **upto 5 meters**. Show the length of **2 meters, 6 decimeters and 5 centimeters** on the scale

Given data : R.F. = 1/50 , maximum length = 5 m and mark 2.65 m

length of scale = R.F X length to measured

$$= 1/50 \times 5 \text{ m}$$

$$= 0.1 \text{ m}$$

$$= 0.1 \times 100 = \mathbf{10 \text{ cm}}$$

Steps for constructing diagonal scale

1. Calculate the length of scale.

length of scale = R.F X length to measured

2. Draw two horizontal lines parallel to each other and some distance apart so that they give appearance as a rule. Join the ends by vertical lines.

(1. Length for horizontal parallel lines = length of scale which is calculated previously 2. assume suitable length)

3. Divide horizontal lines into required number of equal parts and **(depend upon length to measured or read upto)**

4. draw perpendicular lines at these division points.

5. Mark 'zero' at the end of first division.

6. sub-divide only first main part to the left of zero into the required number of sub-division according to unit.

1 m = 10 decimeter

Hence we need to divide first part into 10 equal numbers

7. Sub-divisions are numbered to the right, starting from zero.

8. Mark the given distance on scale

9. name of the scale and R.F. should be mentioned below scale



Conclusion :